THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 12: **Beyond Earth** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A satellite completes 16 orbits in one day (24 hours). The time taken for one orbit is: 1

(a) 90 minutes **(b)** 24 hours **(c)** 120 minutes **(d)** 60 minutes

2. Which two planets can be seen crossing the Sun's disc from Earth? 1

(a) Uranus and Pluto **(b)** Mars and Jupiter
(c) Saturn and Neptune **(d)** Mercury and Venus

3. The Moon is seen as a thin crescent shortly after sunset in the: 1

(a) south **(b)** west **(c)** north **(d)** east

4. Which of the following missions was India's first to reach Mars orbit? 1

(a) Aryabhata **(b)** Chandrayaan-1 **(c)** Aditya **(d)** Mangalyaan

5. The Great Red Spot is a feature of: 1

(a) Jupiter **(b)** Mars **(c)** Venus **(d)** Saturn

6. A planet appears in the evening sky near the western horizon after sunset and is very bright. It is probably: 1

(a) Neptune **(b)** Uranus **(c)** a comet **(d)** Venus

7. Which mission placed the lander near the Moon's south polar region in 2023? 1

(a) Aryabhata **(b)** Chandrayaan-3 **(c)** Mangalyaan **(d)** Chandrayaan-1

8. Which planet spins on its side almost like a rolling ball? 1

(a) Mars **(b)** Earth **(c)** Venus **(d)** Uranus

9. **Assertion (A):** Stars appear to twinkle at night. 1
Reason (R): Their light passes through the moving layers of the air of the Earth.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** There are nine planets in our solar system. 1
Reason (R): Mercury is the planet closest to the Sun.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why can't we see the dark side of the Moon with our eyes? 2

12. Why do countries share space data and run joint space missions? 2

13. The Earth-Moon distance is about 3,80,000 km. On a model where 1 cm stands for 10,000 km, find this distance in cm. 2

14. Explain why a person feels lighter on the Moon. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A signal from Earth takes 5 minutes to reach a spacecraft. If light travels 3 lakh km per second, how far is the spacecraft? 3

16. Why are the outer planets very cold? 3

17. What are the dangers of space debris? 3

OR

Explain why the Sun is called the 'nearest star' and why Proxima Centauri is not. 3

18. Describe the experiment to model the distance of planets using a string. 3

OR

Describe the role of the Indian Space Research Organisation in national development. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Karan watched the Moon over a month. On one night it looked like a thin curved line, a week later it looked half bright, and after about two weeks it appeared as a full bright disc. After a few more days it was gradually shrinking. On one night it was not visible at all. 5
(i) What is the day on which the Moon is not visible called?
(ii) What are the different shapes of the Moon called?
(iii) Does the Moon have its own light?
(iv) Explain why the shape of the bright part seems to change.

20. Evaluate a proposal to settle humans on Mars with advantages and challenges. 5

OR

Explain the importance of the Pole star and the constellations to ancient Indian astronomers. 5

END OF QUESTION PAPER — Total: 40 Marks